AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

Design | Manufacturing | Supply | Installation | Commissioning | AMC

Clean Water Healthy Life Better Tomorrow

WATER SOFTENER

REMOVES HARDNESS (Ca & Mg) AND PROVIDES SOFT WATER FOR INDUSTRIAL, COMMERCIAL & RESIDENTIAL USE

Scale Prevention | Improved Equipment Life | Better Efficiency | Low Maintenance | Reliable Performance

FRP VESSEL
(FIBREGLASS REINFORCED PLASTIC)



FEATURES

- Light weight
- Corrosion resistant
- Non-metallic construction
- Suitable for low to medium capacity
- Easy installation
- Cost effective

MSEP VESSEL
(MILD STEEL EPOXY PAINTED)



FEATURES

- High strength
- Epoxy coated (internal & external)
- Suitable for industrial applications
- Available in custom sizes
- Long service life
- Economical

MSRL VESSEL
(MILD STEEL RUBBER LINED)



FEATURES

- Rubber lining for high corrosion resistance
- Suitable for high hardness and tough water conditions
- Long service life
- Ideal for industrial applications
- Available in custom sizes
- Heavy duty construction

WORKING PRINCIPLE

SERVICE CYCLE
(Hard Water In → Soft Water Out)



REGENERATION CYCLE
(Brine In → Hardness Out)



CAPACITY & VESSEL SIZING (TYPICAL)

Model	Flow Rate (m³/hr)	Vessel Diameter (mm)	Vessel Height (mm)	Resin Quantity (Litres)	Service Flow (m³/hr)	Backwash Flow (m³/hr)
WS-1	1	300	1650	50	1 – 1.2	1.5 – 2.0
WS-2	2	400	1800	100	2 – 2.4	3.0 – 4.0
WS-5	5	600	2000	250	5 – 6	7.5 – 10
WS-10	10	900	2200	500	10 – 12	15 – 20
WS-15	15	1000	2400	750	15 – 18	22 – 30
WS-20	20	1200	2400	1000	20 – 24	30 – 40
WS-30	30	1400	2600	1500	30 – 36	45 – 60
WS-50	50	1600	2600	2500	50 – 60	75 – 100

Note: Above sizes are indicative. Final sizing depends on raw water analysis, OBR, flow rate and operating conditions.

DESIGN BASIS & OPERATING PARAMETERS

Parameter	Typical Design Value
Inlet Hardness (as $CaCO_3$)	50 – 1000 mg/L (design based)
Outlet Hardness	< 5 – 10 mg/L (as $CaCO_3$)
Operating Mode	Continuous / Automatic
Operating Pressure	2.0 – 6.0 kg/cm²
Operating Temperature	5 – 45 °C
Resin Type	Strong Acid Cation (Na^+ form)
Exchange Capacity	≥ 1.8 – 2.0 eq/L (Resin type 220 Na)
Regeneration Chemical	Common Salt (NaCl)
Salt Dose	100 – 150 g per litre of resin
OBR (Between Regeneration)	2 – 12 Hours (based on hardness)

INLET & OUTLET WATER PARAMETERS (TYPICAL)

Parameter	Inlet (Typical)	Outlet (Typical)
Total Hardness (as $CaCO_3$)	50 – 1000 mg/L	< 5 – 10 mg/L
Calcium Hardness	20 – 400 mg/L	< 2 – 5 mg/L
Magnesium Hardness	10 – 300 mg/L	< 2 – 5 mg/L
TDS	No significant change	No significant change
pH	6.5 – 8.5	6.5 – 8.5
Turbidity	< 5 NTU	< 5 NTU
Iron (Fe)	< 0.3 mg/L	< 0.3 mg/L
Oil & Grease	Absent	Absent

Note: Pretreatment (PSF/ACF) is recommended to remove suspended solids, iron & organics.

REGENERATION SEQUENCE

1. Service (Normal Operation)
2. Backwash (5 – 10 min)
3. Brine Draw / Slow Rinse (30 – 60 min)
4. Fast Rinse (10 – 15 min)
5. Return to Service

OBR (OPERATING HOURS BETWEEN REGENERATION) – DESIGN FORMULA

$$\text{Resin Quantity (L)} = \frac{\text{Flow (m}^3\text{/hr)} \times \text{Hardness (mg/L as } CaCO_3\text{)} \times \text{OBR (hrs)}}{1000 \times \text{Exchange Capacity (eq/L)}}$$

Where:

- Flow = service flow rate (m³/hr)
- Hardness = inlet hardness (mg/L as $CaCO_3$)
- OBR = desired operating hours between regeneration (hrs)
- Exchange Capacity = 1.8 – 2.0 eq/L (for 220 Na resin)

SALT & WATER CONSUMPTION (TYPICAL)

Parameter	Typical Value
Salt Dose	100 – 150 g/L of resin
Salt Consumption per Regeneration	(Resin in L × 0.1 – 0.15) kg
Water for Regeneration	2 – 4 m³ per m³ of resin
Backwash Water	5 – 8 m³ per m³ of resin
Slow Rinse Water	1 – 2 m³ per m³ of resin
Fast Rinse Water	2 – 4 m³ per m³ of resin

APPLICATIONS

- Boiler Feed Water Treatment
- Cooling Tower Make-up
- Process Water for Industries
- Hotels, Hospitals & Commercial Buildings
- Pharmaceutical & Food Industry
- Textile, Paper & Chemical Industry
- Pre-RO Softening
- Residential & Institutional Use

ACTUAL SITE INSTALLATION



AQUEN FLUID TECHNOLOGIES LLP

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WTP | STP | ETP | ZLD | FABRICATION | SPARES | AMC | RETROFIT | AUTOMATION

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Site 1 – FRP Vessels (MGF + Softener)



Site 2 – MSRL Vessels (MGF + Softener)



Site 3 – FRP Vessels with Pump Skid



Site 4 – MSEP Vessels with Regeneration Tank



Site 5 – Indoor Installation (FRP)



Site 6 – MSRL Vessels with Brine Tank



Butterfly Valve Frontal Piping with Regeneration Tank

